

# STATISTICAL STRUCTURED PREDICTION

Part I: **José Miguel Benedí** e-mail: jmbenedi@prhlt.upv.es

**Joan Andreu Sánchez** e-mail: jandreu@prhlt.upv.es

Part II: **Enrique Vidal** e-mail: evidal@prhlt.upv.es



Statistical Structured Prediction

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J.M. Benedí (2022-2023)

Statistical Structured Prediction 1

## STATISTICAL STRUCTURED PREDICTION

|                                 |                                                                                                                                       |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| <b>Técnicas Fundamentales</b>   | <b>R.F. y Aprendizaje Computacional</b><br><b>Lingüística Computacional</b>                                                           |
| <b>Reconocimiento de Formas</b> | <b>Predicción Estructurada Estadística</b><br><b>Redes Neuronales Artificiales</b><br><b>Aplicaciones de Reconocimiento de Formas</b> |
| <b>Tecnologías del Lenguaje</b> | <b>Traducción Automática</b><br><b>Reconocimiento Automático del Habla</b><br><b>Aplicaciones de la Lingüística Computacional</b>     |
| <b>Técnicas Complementarias</b> | <b>Aprendizaje Automático Avanzado</b><br><b>Reconocimiento de Escritura</b><br><b>Biometría</b><br><b>Visión por Computador</b>      |

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Statistical Structured Prediction 2

## STATISTICAL STRUCTURED PREDICTION

### Focus of Course

- Give students an overview of the different topics that allow them to understand the basic concepts:
  - What is structured prediction?
  - What problems will we try to address?
  - How to make predictions and learn the models in structured output spaces?
  - What are the different computational challenges for structured prediction?
- We will deal with the rigorous design of algorithms and make intensive use of mathematics, but nothing too hard.



*"There is nothing more practical than a good theory".*

Kurt Lewin

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Statistical Structured Prediction 3

## Syllabus

Part I: Statistical Structured Prediction for structured output data

1. Introduction
2. Models for Statistical Structured Prediction
3. Making Prediction: Decoding and Inference
4. Model Parameter Estimation

Part II: Fundamentals of Multimodal Interaction in Pattern Recognition

## Bibliography of Part I

### Pattern Recognition and Machine Learning

- Christopher M. Bishop: **Pattern Recognition and Machine Learning**. Springer.
- Kevin P. Murphy: **Machine Learning: A Probabilistic Perspective**. The MIT Press.
- Daphne Koller and Nir Friedman: **Probabilistic Graphical Models: Principles and Techniques**. The MIT Press.
- Richard O. Duda, Peter E. Hart and David G. Stork: **Pattern Classification** (2nd ed.). Wiley Interscience.

## Statistical Models for Natural Language Processing

- Christopher D. Manning and Hinrich Schütze: **Foundations of Statistical Natural Language Processing**. The MIT Press.
- Noah A. Smith: **Linguistic Structure Prediction**. Morgan & Claypool.
- Daniel Jurafsky and James H. Martin: **Speech and Language Processing**. Prentice Hall (2<sup>a</sup> ed).

## Lectures

Monday 15:00 - 18:00 (10 sessions of 3 hours)

|                |                     |                            |
|----------------|---------------------|----------------------------|
| <b>Part I</b>  | <b>J.M. Benedí</b>  | November 07, first session |
|                | <b>J.A. Sánchez</b> | November 28, first session |
| <b>Part II</b> | <b>E.Vidal</b>      | January 9, first session   |

## Assessment

- 3 Collections of selected exercises (theoretical and practical)
- 1 Multiple-choice test (Monday, February 13, 2023)

|    | delivery       | deadline       |
|----|----------------|----------------|
| Q1 | 28 - 11 - 2022 | 16 - 12 - 2022 |

## Tutoring

Tutoring available by previous appointment

José Miguel Benedí <jmbenedi@prhlt.upv.es> (office: 1D13)

## 1. Introduction

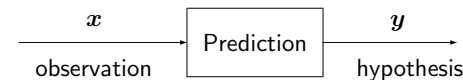
### 1.1. Structured Prediction: motivation and overview

### 1.2. Predicting sequences

- Hidden Markov Models

### 1.3. Syntactic Parsing

- Probabilistic Context-Free Grammars



- Input observation;  $x \in \mathcal{X}$  can be any kind of object.
- Output hypothesis;  $y \in \mathcal{Y}$  is a real number:  $\mathcal{Y} = \{1, \dots, K\}$  or  $y \in \mathbb{R}$ .
- (Non-structured) prediction function;  $f : \mathcal{X} \rightarrow \mathbb{R}$  assigns a hypothesis  $y = f(x)$  to each entry  $x$ .
- Binary classification:  $y \in \{-1, 1\}$ .
- Multiclass classification:  $y \in \{1, \dots, K\}$ .
- Regression:  $y \in \mathbb{R}$ .

## Algorithms for predicting non-structured output data

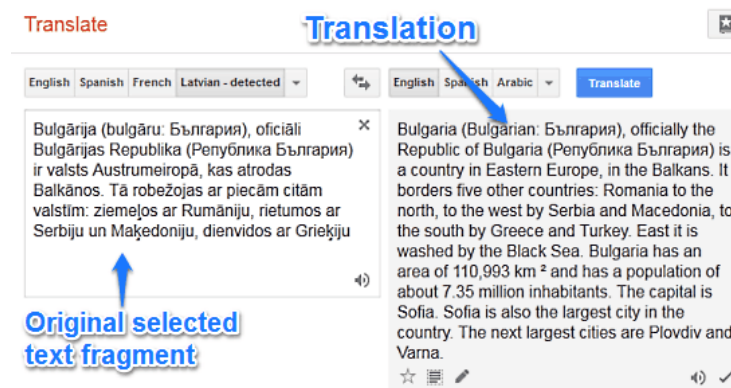
|                         |                              |
|-------------------------|------------------------------|
| Naive Bayes classifier  | Regression                   |
| Logistic Regression     | Fisher's linear discriminant |
| Perceptron algorithm    | K-Nearest Neighbor           |
| Support Vector Machines | Random Forests               |
| Neural Networks         | ...                          |

What if the space of outputs is much larger and more structured?  
Say trees, or in general, graphs.

## Machine Translation

Input: Text sequences

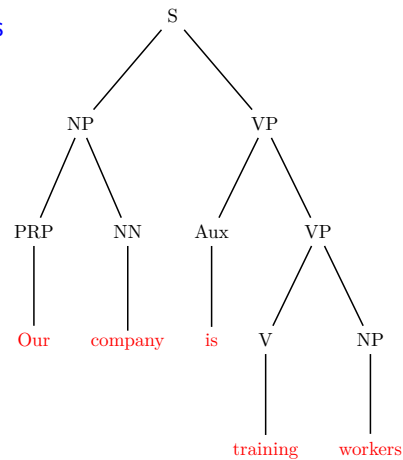
Output: Text sequences



## Syntactic Parsing

**Input:** Text sequences

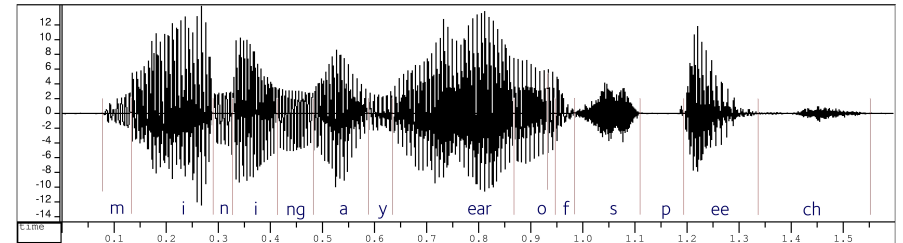
**Output:** Parse trees



## Automatic Speech Recognition

**Input:** Speech signals

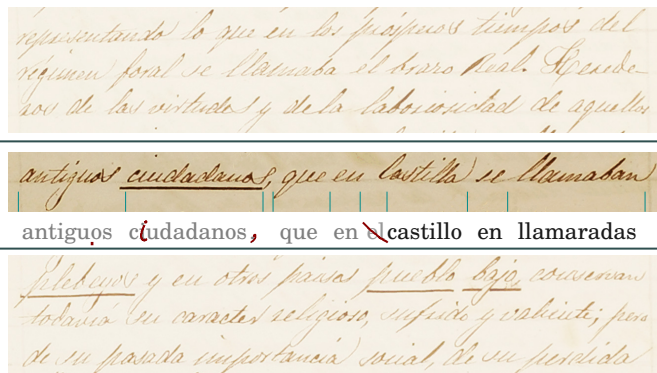
**Output:** Transcribed text sequences



## Handwriting Text Recognition

**Input:** Images

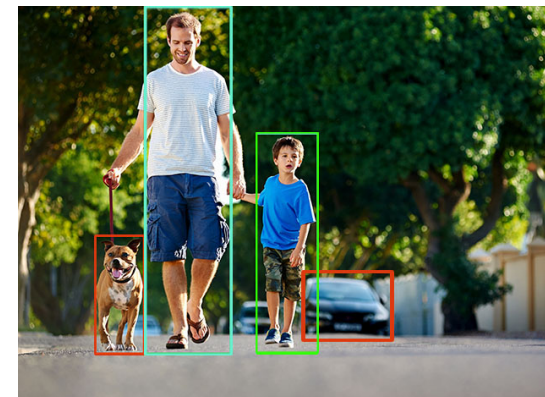
**Output:** Transcribed text sequences



## Scene Analysis

**Input:** Images

**Output:** Scene layout graphs

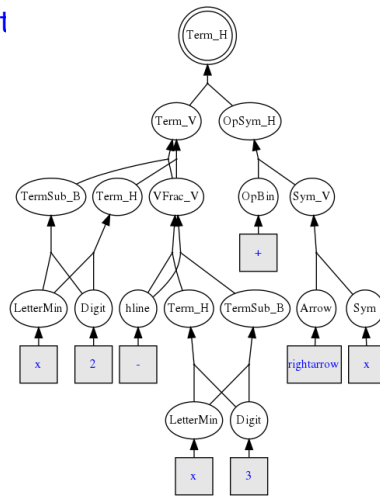


## Mathematical Expressions Recognit

Input: Images

Output: Hypergrphs

$$\frac{x_2}{x_3} + \vec{x}$$



⇒ Mathematical Expression Recognition

## Outputs

## ➤ Natural Language Processing:

- Part-of-Speech tagging (sentences)
- Parsing (parse trees)
- Machine Translation (sentences or hypergraphs)
- Information Extraction (sentences)

## ➤ Image Processing:

- Visual Scene Analysis (sentences or relationship graphs)
- Handwritten Text Recognition (sentences or word graphs)

## ➤ Speech Processing:

- Automatic transcription (sentences or word graphs)
- Text-to-Speech (audio signal)

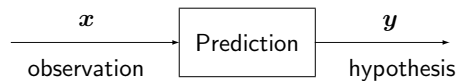
## ➤ Bioinformatics:

- Protein Structure Prediction (graphs)

## ➤ Robotics:

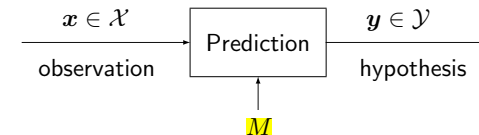
- Planning (sequence of actions)

## STRUCTURED PREDICTION



- Input observation;  $x \in \mathcal{X}$  can be any kind of objects
- Output hypothesis;  $y \in \mathcal{Y}$  is a complex structured object.
- (Structured) prediction function;  $f: \mathcal{X} \rightarrow \mathcal{Y}$  which assigns a hypothesis  $y = f(x)$  to each entry  $x$ .
  - $y$  can be a sequence,
  - $y$  can be a parse tree,
  - $y$  can be a graph.

## STRUCTURED PREDICTION: SEARCH SPACE

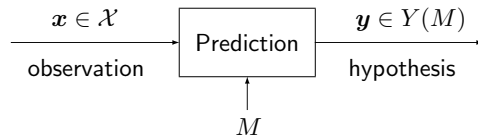


- Typically,  $\mathcal{Y}$  is potentially infinite  $\implies f(x) = y \in \mathcal{Y}$  will often be intractable.
- (Structural) Solution: We assume a specific structure in  $\mathcal{Y}$ , and we define a (finite) model  $M$  that allows us to characterize this structure:  $Y(M) \subseteq \mathcal{Y}$

$$f(x) = y \in Y(M)$$

Depending on  $M$ , we can find polynomial solutions.

## STRUCTURED PREDICTION: UNCERTAINTY



### > Real life

- > **Fragility**: Lack of robustness due to an imperfect representation of input objects  
[ Uncertainty regarding representations of inputs ]
- > **Ambiguity**: Several interpretations (hypothesis) are plausible due to variability and difficulty of the task. [ Uncertainty regarding outputs ]
- > **Incompleteness**: Inadequate models due to incomplete and insufficient knowledge  
[ uncertainty ! ]

### > (Statistical) Solution Based on statistical decision theory $f : \mathcal{X} \rightarrow Y(M)$

$$\hat{y} = f(x) = \arg \max_{y \in Y(M)} P(y | x)$$

## STATISTICAL STRUCTURED PREDICTION: TAXONOMY OF MODELS

**Discriminative models** (conditional probability)  $P(y | x)$

$$f(x) = \arg \max_{y \in Y(M)} P(y | x)$$

**Generative models** (joint probability)  $P(x, y)$

$$f(x) = \arg \max_{y \in Y(M)} P(x, y) = \arg \max_{y \in Y(M)} P(x | y) \cdot P(y)$$

## STATISTICAL STRUCTURED PREDICTION

### Three Key Components

- > **Statistical model**: used to replace the unknown real models  
 $P(y | x)$  or  $P(x | y)$  and  $P(y)$  vs  $P_\theta(y | x)$  or  $P_\theta(x | y)$  and  $P_\theta(y)$   
Discriminant Function, Neural networks, Gaussian mixtures, models with hidden variables (HMM, alignment models), log-linear models, etc.
- > **Decision rule (search or decoding)**: Sometimes hard !  
Dynamic Programming, A\* search, etc.
- > **Training criterion** to learn the unknown parameters  $\theta$  from training data  
Maximum Likelihood Estimation, Maximum a Posterior Estimation, Minimum Classification error, etc.

Specific algorithms depend on statistical models

## SEQUENCES

### > Sequences of states

- Text is a sequence of words or even letters
- A spoken utterance is a sequence of parameter vectors
- A video is a sequence of frames
- ...

### > Goal (for now): Define probability distributions over sequences

$$x_1 \ x_2 \ \dots \ x_n \implies P(x_1, x_2, \dots, x_n)$$

### > A naive history-based model

$$P(x_1, x_2, \dots, x_n) = \prod_{i=1}^n P(x_i | x_1 \ x_2, \dots, x_{i-1})$$

## SEQUENCES

➤ **Solution:** Limit the history

### First-order Markov chain assumption

The state of the process at any time is independent of the complete sequence history given the previous state

$$P(x_i | x_1 x_2, \dots, x_{i-1}) = P(x_i | x_{i-1})$$

Therefore, this allows us to simplify

$$P(x_1, x_2, \dots, x_n) = \prod_{i=1}^n P(x_i | x_{i-1})$$

Defined by two sets of probabilities:

- The initial state distribution  $P(x_1) \equiv P(x_1 | \#)$
- The state transition distribution  $P(x_i | x_{i-1})$

## HIDDEN MARKOV MODEL

➤ **First-order Markov Model:**

- States follow a Markov chain
- **Each state is an observation**

➤ **Hidden Markov Model:**

- States follow a Markov chain
- **States are not observed**
- Each state **stochastically** emits an observation

## HIDDEN MARKOV MODELS

**Joint model** over states and observations

$$P(x_1, \dots, x_n, y_1, \dots, y_n) = P(x_1, \dots, x_n | y_1, \dots, y_n) P(y_1, \dots, y_n)$$

$$= \prod_{i=1}^n P(x_i | x_1, \dots, x_{i-1}, y_1, \dots, y_i) \prod_{i=1}^n P(y_i | y_1, \dots, y_{i-1})$$

### Generative models: Hidden Markov Models

**Assumptions:** 1) each state depends only on its immediate predecessor;  
2) each observation depends only on the current state.

- Observation distribution  $P(x_i | y_i) \equiv e(x_i | y_i)$
- Transition distribution  $P(y_i | y_{i-1}) \equiv q(y_i | y_{i-1})$
- Distribution over initial states  $P(y_1 | \#) \equiv q(y_1 | y_0)$

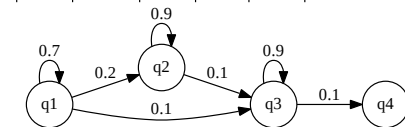
$$P(x_1, \dots, x_n, y_1, \dots, y_n) = \prod_{i=1}^n q(y_i | y_{i-1}) e(x_i | y_i)$$

## HIDDEN MARKOV MODELS

**Example:** Given a HMM  $(Q, \Sigma, \theta)$ , where:  $Q = \{1, 2, 3, 4\}$  is the set of states,  $\Sigma = \{a, b\}$  is the set of output symbols, and  $\theta = (q, e)$  is the parameter vector:

**Symbol emission probabilities**

|   |     |   |     |   |     |
|---|-----|---|-----|---|-----|
| a | 0.9 | a | 0.1 | a | 0.9 |
| b | 0.1 | b | 0.9 | b | 0.1 |



**State transition probabilities**

**Initial state probabilities**

$$q(q_1 | \#) = 1; \quad q(q_2 | \#) = q(q_3 | \#) = q(q_4 | \#) = 0$$

$$P_\theta(aba, q_1 q_2 q_3 q_4) = q(q_1 | \#) e(a | q_1) q(q_2 | q_1) e(b | q_2) q(q_3 | q_2) e(a | q_3) q(q_4 | q_3) = 1,458 \cdot 10^{-3}$$

$$P_\theta(aba, q_1 q_1 q_3 q_4) = q(q_1 | \#) e(a | q_1) q(q_1 | q_1) e(b | q_1) q(q_3 | q_1) e(a | q_3) q(q_4 | q_3) = 0,567 \cdot 10^{-3}$$

$$P_\theta(aba, q_1 q_3 q_3 q_4) = q(q_1 | \#) e(a | q_1) q(q_3 | q_1) e(b | q_3) q(q_3 | q_3) e(a | q_3) q(q_4 | q_3) = 0,729 \cdot 10^{-3}$$

$$P_\theta(aba) = 2,754 \cdot 10^{-3}$$

## Three key components for HMMs

[Rabiner, 1999]

1. Given an observation sequence  $x_1, \dots, x_n$  and a model  $(Q, \Sigma, \theta)$ , how to efficiently calculate the probability of the observation?
2. Given an observation sequence  $x_1, \dots, x_n$  and a model  $(Q, \Sigma, \theta)$ , how to efficiently calculate the most probable state sequence?
3. How to calculate the parameters  $\theta$  of the model  $(Q, \Sigma, \theta)$  from a set of observations?

## Generalizations

- > High-order HMMs: Trigram HMMs

$$P(x_1, \dots, x_n, y_1, \dots, y_n) = \prod_{i=1}^n q(y_i | y_{i-2}, y_{i-1}) e(x_i | y_i)$$

- > Continuous HMMs (gaussian mixture densities)

If the observations are continuous vectors  $\mathbf{x}$ , the symbol emission probabilities  $e(x|j)$  can be replaced by the continuous density  $e(\mathbf{x}|j)$ , such that  $\forall j \in Q$ :

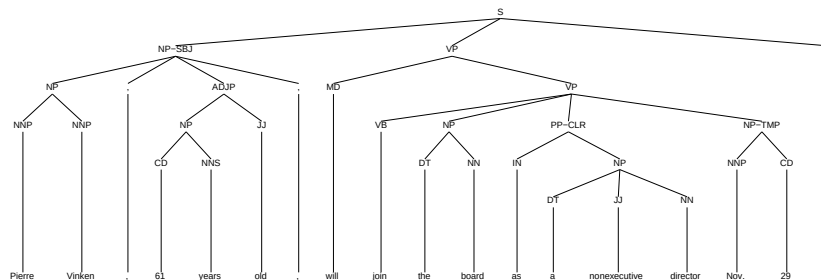
$$e(\mathbf{x}|j) = \sum_k c_{jk} \mathcal{N}(\mathbf{x}, \mu_{jk}, \sigma_{jk})$$

where  $c_{jk}$  is the mixture weight,  $\mathcal{N}$  is the normal density, and  $\mu_{jk}$  and  $\sigma_{jk}$  are the mean vector, and the covariance matrix associated with state  $j$  and mixture  $k$ .

## SYNTACTIC PARSING

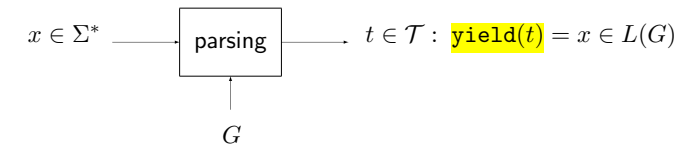
**Example:** “Pierre Vinken, 61 years old, will join the board as a nonexecutive director Nov. 29.” [Marcus et al., 1993]

((S (NP-SBJ (NP (NNP Pierre) (NNP Vinken)) (.,) (ADJP (NP (CD 61) (NNS years)) (JJ old)) (.,)) (VP (MD will) (VP (VB join) (NP (DT the) (NN board)) (PP-CLR (IN as) (NP (DT a) (JJ nonexecutive) (NN director)))) (NP-TMP (NNP Nov.) (CD 29)))) (..)))



## SYNTACTIC PARSING

- > Parsing as a search problem



- > Grammatical models: reasons for use

- > Grammatical models are the simplest and most natural model for tree structures
- > Formal (mathematical) framework is well known
- > Compact models (small number of free parameters)
- > Good behavior against the problem of ambiguity
- > Context-free grammatical models represent well long-term dependencies of syntactic and semantic constraints of Natural Language

## CONTEXT-FREE GRAMMARS

### Context-Free Grammars $G = (\Sigma, N, S, \mathcal{P})$

$\Sigma$  a set of **terminal symbols**

$N$  a set of **non-terminals symbols** (or variables):  $N \cap \Sigma = \emptyset$

$S$  a distinguished **start symbol**:  $S \in N$

$\mathcal{P}$  a set of **rules** (or productions):  $(A \rightarrow \alpha) \in \mathcal{P}; A \in N; \alpha \in (N \cup \Sigma)^*$

### Direct derivation: $\delta A \gamma$ **directly derives** $\delta \alpha \gamma$ or

$\delta A \gamma \Rightarrow \delta \alpha \gamma$  **iff**  $\exists (A \rightarrow \alpha) \in \mathcal{P}; \delta, \gamma \in (N \cup \Sigma)^*$

### Derivation: $\alpha$ **derives** $\beta$ or

$\alpha \Rightarrow^* \beta$  **iff**  $\exists \alpha_0, \dots, \alpha_m \in (N \cup \Sigma)^*: \alpha = \alpha_0 \Rightarrow \alpha_1 \Rightarrow \dots \Rightarrow \alpha_{m-1} \Rightarrow \alpha_m = \beta$

### Language generated by a grammar: $L(G) = \{x \mid x \in \Sigma^*: S \Rightarrow^+ x\}$

**Theorem**  $x \in L(G)$  **iff**  $S \Rightarrow^+ x$  **iff**  $\exists t \in \mathcal{T}: \text{yield}(t) = x$

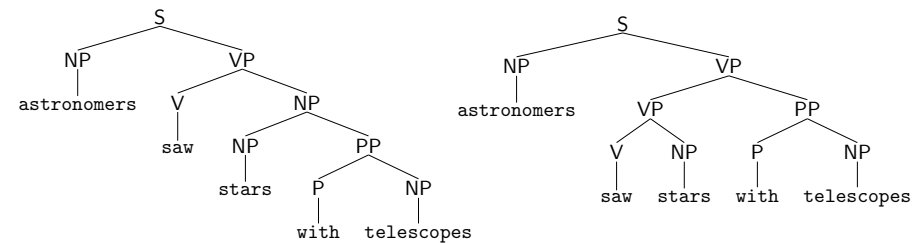
## CONTEXT-FREE GRAMMARS: EXAMPLE

### Example: A simple Context-Free Grammars

[Manning and Schütze, 2002]

|                                      |                                      |                                            |                                           |
|--------------------------------------|--------------------------------------|--------------------------------------------|-------------------------------------------|
| $S \rightarrow \text{NP VP}$         | $\text{VP} \rightarrow \text{V NP}$  | $\text{V} \rightarrow \text{saw}$          | $\text{NP} \rightarrow \text{saw}$        |
| $\text{NP} \rightarrow \text{NP PP}$ | $\text{VP} \rightarrow \text{VP PP}$ | $\text{NP} \rightarrow \text{astronomers}$ | $\text{NP} \rightarrow \text{stars}$      |
| $\text{PP} \rightarrow \text{P NP}$  | $\text{P} \rightarrow \text{with}$   | $\text{NP} \rightarrow \text{ears}$        | $\text{NP} \rightarrow \text{telescopes}$ |

$S \Rightarrow \text{NP VP} \Rightarrow \text{astronomers VP} \Rightarrow \text{astronomers V NP} \Rightarrow \text{astronomers saw NP} \Rightarrow$   
 $\text{astronomers saw NP PP} \Rightarrow \text{astronomers saw stars PP} \Rightarrow \text{astronomers saw stars P NP}$   
 $\Rightarrow \text{astronomers saw stars with NP} \Rightarrow \text{astronomers saw stars with telescopes}$



## CKY PARSING



### Algorithm 1: Cocke-Kasami-Younger

**Input:**  $G = (\Sigma, N, S, \mathcal{P})$  in FNC and

$\mathbf{x} = x_1 \dots x_T \in \Sigma^*$

**Output:** Parsing table  $t[i, j]$  ( $1 \leq i, j \leq T$ );

$A \in t[i, i + l]$  **iff**  $A \Rightarrow^* x_{i+1} \dots x_{i+l}$

**for**  $i: 0 \dots T - 1$  **do**

$t[i, i + 1] = t[i, i + 1] \cup \{A : (A \rightarrow b) \in \mathcal{P}; b = x_{i+1}\}$

**for**  $l: 2 \dots T$  **do**

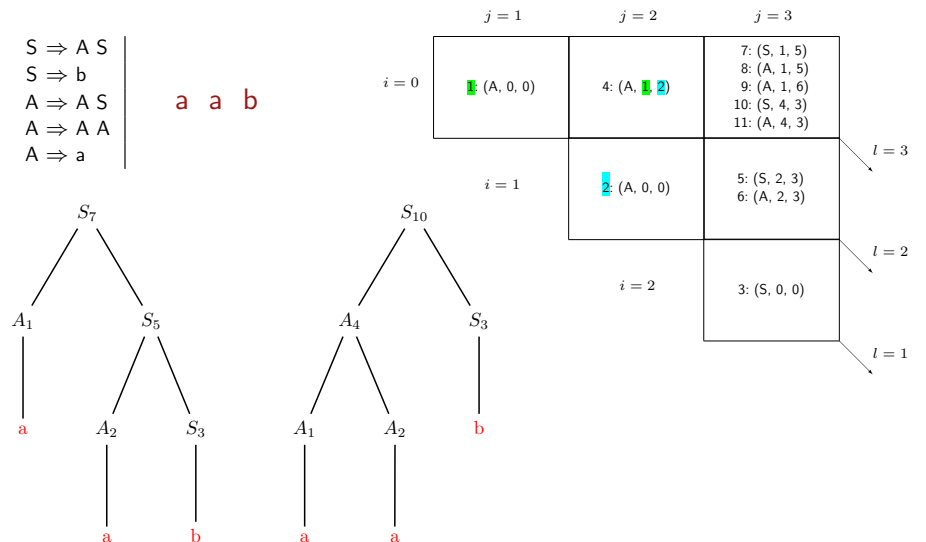
**for**  $i: 0 \dots T - l$  **do**

**for**  $k: 1 \dots l - 1$  **do**

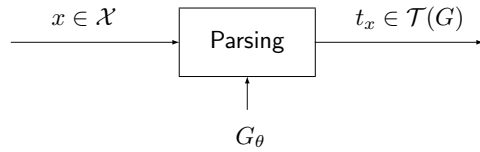
$t[i, i + l] = t[i, i + l] \cup \{A : (A \rightarrow BC) \in \mathcal{P};$   
        $B \in t[i, i + k]; C \in t[i + k, i + l]\}$ ;

**if**  $S \in t[0, T]$  **then**  $x \in L(G)$  **else**  $x \notin L(G)$ ;

## CKY PARSING: EXAMPLE



## STATISTICAL STRUCTURED PREDICTION: PARSING



$$t_x \in \mathcal{T}(G) : \text{yield}(t) = x \quad \text{iff} \quad S \stackrel{\pm}{\Rightarrow} x \quad \text{iff} \quad x \in L(G)$$

### ➤ Real life [ Uncertainty ]

- **Fragility:** Lack of robustness due to imperfect representation of input objects
- **incompleteness:** Inadequate models due to incomplete and insufficient knowledge
- **Ambiguity:** Several possible interpretations due to variability of the task.

### ➤ (Statistical) Solution based on statistical decision theory: **Probabilistic models**

$$x \in L(G) \quad \Rightarrow \quad P(x; G_\theta) \equiv P(x; \theta) \equiv P_\theta(x)$$

where  $\theta$  is the parameter vector of the probabilistic model  $G_\theta$

## PROBABILISTIC LANGUAGES

➤ **Language**  $L \subseteq \Sigma^*$ , given an alphabet  $\Sigma$

➤ **Language generated by a grammar**  $L(G) \subseteq \Sigma^*$ ,  
given an alphabet  $\Sigma$  and a grammar  $G = (\Sigma, N, S, \mathcal{P})$

$$L(G) = \{x \mid x \in \Sigma^* : S \stackrel{\pm}{\Rightarrow} x\}$$

➤ **Probabilistic language**  $(L, \phi)$ , given an alphabet  $\Sigma$

- $L \subseteq \Sigma^*$  characteristic language
- $\phi : \Sigma^* \rightarrow [0, 1]$  computable **probabilistic** function:

$$\text{i) } x \notin L \Rightarrow \phi(x) = 0 \quad \forall x \in \Sigma^*$$

$$\text{ii) } x \in L \Rightarrow 0 < \phi(x) \leq 1 \quad \forall x \in \Sigma^*$$

$$\text{iii) } \sum_{x \in L} \phi(x) = 1$$

## PROBABILISTIC LANGUAGES

### ➤ Example [Booth-Thompson,73]

Given the alphabet  $\Sigma = \{a, b\}$ , the following language is defined:

$$L = \{a^n b^n \mid n \geq 0\}, \quad \text{where } \phi(x) = 0, \quad \forall x \notin L \quad \text{and} \quad \phi(a^n b^n) = \frac{1}{e \cdot n!}$$

$$\sum_{x \in L} \phi(x) = \sum_{0 \leq n \leq \infty} \frac{1}{e \cdot n!} = \frac{1}{e} \sum_{0 \leq n \leq \infty} \frac{1}{n!} = \frac{1}{e} e = 1$$

### ➤ **Theorem** [Wetherell,80]

Let  $(L, \phi)$  be an infinite probabilistic language, then for each  $\epsilon > 0$

there exists an  $n \geq 0$ , such as:

$$|x| \geq n \Rightarrow \epsilon > \phi(x)$$

## PROBABILISTIC CONTEXT-FREE GRAMMARS

➤ **Probabilistic Context-Free Grammars:**  $G_\theta = (G, P)$

- $G = (\Sigma, N, S, \mathcal{P})$  characteristic grammar
- $P : \mathcal{P} \rightarrow [0, 1]$  probabilities of rules:

$$\forall A_i \in N \quad P(A_i \rightarrow \alpha_j) \equiv P(r_{ij}) \equiv P(A_i \rightarrow \alpha_j \mid A_i) \equiv P(\alpha_j \mid A_i)$$

where  $n_i$  is the number of rules with  $A_i$  in the left side of rule:

$$\sum_{1 \leq j \leq n_i} P(\alpha_j \mid A_i) = 1$$

## PROBABILISTIC CONTEXT-FREE GRAMMARS

### ➤ Probabilistic derivation

Given a sequence of stochastic events:

$$S = \alpha_0 \xRightarrow{r_1} \alpha_1 \xRightarrow{r_2} \alpha_2 \cdots \alpha_{m-1} \xRightarrow{r_m} \alpha_m = x$$

the probability of  $x$  being generated by  $G_\theta = (G, P)$  from the rule sequence

$t_x = r_1 \cdots r_m$ , is:

$$P_\theta(x, t_x) = P(r_1) \cdot P(r_2 | r_1) \cdots P(r_m | r_1 \cdots r_{m-1})$$

➤ **problem:** computation of the probabilities

➤ **restriction:**  $P(r_j | r_1 \cdots r_{j-1}) = P(r_j)$

$$P_\theta(x, t_x) = \prod_{j=1 \cdots m} P(r_j)$$

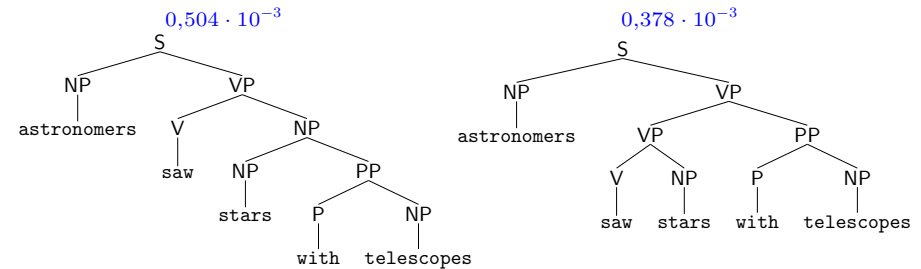
## PROBABILISTIC CONTEXT-FREE GRAMMARS: EXAMPLE

Example: A simple Context-Free Grammars

[Manning and Schütze, 2002]

|                          |     |                             |     |                                     |      |                                    |      |
|--------------------------|-----|-----------------------------|-----|-------------------------------------|------|------------------------------------|------|
| $S \rightarrow NP \ VP$  | 1,0 | $VP \rightarrow V \ NP$     | 0,7 | $V \rightarrow \text{saw}$          | 1,0  | $NP \rightarrow \text{saw}$        | 0,04 |
| $NP \rightarrow NP \ PP$ | 0,4 | $VP \rightarrow VP \ PP$    | 0,3 | $NP \rightarrow \text{astronomers}$ | 0,1  | $NP \rightarrow \text{stars}$      | 0,18 |
| $PP \rightarrow P \ NP$  | 1,0 | $P \rightarrow \text{with}$ | 1,0 | $NP \rightarrow \text{ears}$        | 0,18 | $NP \rightarrow \text{telescopes}$ | 0,1  |

$S \xRightarrow{1,0} NP \ VP \xRightarrow{0,1} \text{astronomers} \ VP \xRightarrow{0,7} \text{astronomers} \ V \ NP \xRightarrow{1,0} \text{astronomers} \text{saw} \ NP \xRightarrow{0,4}$   
 $\text{astronomers} \text{saw} \ NP \ PP \xRightarrow{0,18} \text{astronomers} \text{saw} \text{stars} \ PP \xRightarrow{1,0} \text{astronomers} \text{saw} \text{stars} \ P \ NP \xRightarrow{1,0}$   
 $\text{astronomers} \text{saw} \text{stars} \text{with} \ NP \xRightarrow{0,1} \text{astronomers} \text{saw} \text{stars} \text{with} \text{telescopes} = 0,504 \cdot 10^{-3}$



## PROBABILISTIC CONTEXT-FREE GRAMMARS

### ➤ Probability of a parse tree

$$P_\theta(x, t_x) = \prod_{j=1 \cdots m} P(r_j)$$

### ➤ Probability of a string

$$P_\theta(x) = \sum_{t_x \in \mathcal{T}_x} P_\theta(x, t_x)$$

### ➤ Probability of the best parse tree

$$\widehat{P}_\theta(x) = \max_{t_x \in \mathcal{T}_x} P_\theta(x, t_x)$$

### ➤ Language generated by a probabilistic grammar

$$L(G_\theta) = \{x \in L(G) \mid P_\theta(x) > 0\}$$

## PROBABILISTIC CONTEXT-FREE GRAMMARS: PROPERTIES

### ➤ Consistent grammars

A probabilistic grammar  $G_\theta = (G, P)$  is consistent **iff**:

$$\sum_{x \in L(G)} P_\theta(x) = 1$$

### ➤ Theorem [Booth-Thompson,73]

There exist probabilistic languages  $(L, \phi)$  that can not be generated by a probabilistic grammar  $G_\theta = (G, P)$

### ➤ Example.- Let $L = \{a^n b^n \mid n \geq 0\}$ be a probabilistic language:

$$\phi(a^n b^n) = \frac{1}{e \cdot n!}$$

There is not any  $G_\theta$  such that  $\phi(x) = P_\theta(x) \quad \forall x \in L$

## COURSE COVERAGE

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| Problems       | Models         |                                                                                                                                                                        |
|----------------|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Non-structured | Generative     | Naive Bayes classifier                                                                                                                                                 |
|                | Discriminative | Logistic Regression<br>Perceptron algorithm<br>Support Vector Machines<br>Neural networks                                                                              |
| Structured     | Generative     | <a href="#">Hidden Markov Models</a><br><a href="#">Probabilistic Context-Free Grammars</a>                                                                            |
|                | Discriminative | <a href="#">Conditional Random Fields</a><br><a href="#">Structured Perceptron</a><br>Structured Support Vector Machines<br>(Encoder-Decoder) Recurrent Neural Network |